Topic 7 – Safe working practices

7.1 Safe working practices and risk assessment

Spec: Understanding safe working practices for yourself and others when designing and making, including when selecting and safely using machinery, equipment and tools in order to ensure safe working environments; Understanding the need for risk assessments – identification of potential hazards, identification of people at risk, evaluation of risks, implement control measures, recording and storing of risk assessment documentation

Key facts

• **HSE five steps to risk assessment?** 1 Identify hazards; 2 Decide who might be harmed and how; 3 Evaluate risks and decide on precautions (control measures); 4 Record findings and implement; 5 Review and update.
• **Hazard vs risk?** Hazard = something that could cause harm. Risk = the chance it will cause harm and how severe it would be.
• **Hierarchy of control order?** Eliminate, substitute, engineering controls (guards, extraction), administrative (training, signs, procedures), PPE.
• **Two parts of a risk assessment other than the level of risk?** Identifying hazards; identifying people at risk; control measures; responsibilities; review dates.
• **Safe working practices for a bandsaw (excluding PPE)?** Lower blade guard to just above work, check blade tension, use push stick, keep fingers clear, tie back hair/no loose clothing, know emergency stop, support work flat on table, extraction on, only trained users.
• **Who could be at risk in a workshop?** Operator, other students/workers, teachers/technicians, visitors, cleaners, maintenance staff.
• **Why record and store risk assessments?** Legal requirement (5+ employees), evidence for HSE/insurance, communicate hazards and controls to staff, allows review after incidents or changes.

Equipment	Key hazards	Key controls
Bandsaw / scroll saw	Cuts, entanglement	Guard just above work, push stick, blade tension checked, emergency stop
Pillar drill	Entanglement, swarf, work spinning	Clamp work in machine vice, chuck guard, remove chuck key, goggles
Centre lathe	Entanglement, swarf, flying chuck key	Guards, remove chuck key, goggles, no gloves near rotating work
Brazing / casting / welding	Burns, fumes, UV, molten metal	Leather apron and gauntlets, face shield, sand tray, extraction, screens
Laser cutter	Fire, fumes, eye damage	Never unattended, extraction, approved materials only, lid interlock
Sanders / routers	Dust, noise, abrasion	Extraction, dust mask, ear defenders, guards
Solvents, resins, paints	Toxic fumes (VOCs), flammability, skin irritation	COSHH assessment, ventilation/spray booth, gloves, goggles, fire cabinet

Likelihood (1–5)	Severity (1–5)
1 highly unlikely · 2 remote · 3 possible · 4 likely · 5 frequent/almost certain	1 negligible · 2 minor injury · 3 injury needing treatment · 4 major injury · 5 fatality

Key terms

Hazard	Anything with the potential to cause harm – e.g. a rotating blade, molten metal, solvent fumes, trailing cable.
Risk	The likelihood that harm from a hazard will actually happen, combined with how severe the harm would be.
Risk assessment	A careful, recorded examination of what could cause harm, who could be harmed and whether enough precautions have been taken.
Control measure	An action that removes a hazard or reduces the risk – e.g. guards, extraction, training, PPE.
Hierarchy of control	Elimination → substitution → engineering controls → administrative controls → PPE (last resort).
Residual risk	The risk remaining after control measures are in place.
Risk rating	Likelihood × severity (e.g. on a 1–5 scale each) giving a score or position on a risk graph/matrix.

Exam tip (2024 Q6a): If the question says "other than PPE", PPE answers score zero. In 2024 Q6d, "severity" and "likelihood" were given in the stem, so the marks were for **hazards, people at risk, control measures, responsibilities and review dates**.